



I Semester, 2026–2027
Course Handout Part II

Date: 01 August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : PHY F416
Course Title : SOFT CONDENSED MATTER PHYSICS
Instructor-in-Charge : Prof Navin Singh

1. Course Description:

Soft matter or soft condensed matter is a subfield of condensed matter comprising a variety of physical systems that are deformed or structurally altered by thermal or mechanical stress of the magnitude of thermal fluctuations. They include liquids, colloids, polymers, foams, gels, granular materials, liquid crystals, and biomolecules. These materials share an essential characteristic feature in that predominant physical behaviors occur at an energy scale comparable with room temperature thermal energy. At these temperatures, quantum aspects are less significant.

In this course, our aim would be to understand the structure, properties, and dynamics of the soft matter. What is the order of forces and times scales that affect the overall behaviour of this class of matter?

2. Scope & Objective of the Course:

- Develop familiarity with the basics of soft condensed matter.
- To develop an understanding about the forces and order of thermal fluctuations in soft matter.
- Why some part of matter is classified as soft matter?

3. Textbooks & References:

1. **T1:** Soft Matter Physics, Masao Doi, Oxford University Press, I edition (2013)
2. **T2:** Soft Condensed Matter, R.A.L. Jones, Oxford University Press, I edition (2002)
3. **R1:** Essentials of Soft Matter Science, Linda S. Hirst, CRC Press (Taylor & Francis), II edition (2020).
4. **R2:** Lecture Notes/ Research Papers relevant to the topics

4. Course Plan:

Module Sessions	Learning Objectives	Refs.	Learning Outcomes
Basics of Soft Matter (1-3)	(i) Why it is called as soft matter? (ii) Concept of thermal energies (iii) How it is different than condensed matter? (iv) Three states of matter: solid, liquid & gas (v) Inter-molecular forces	1.1-1.5 (T1)	The students shall be able to identify the concept of soft matter (or soft condensed matter).
Phase transitions in soft matter (2-7)	(i) Thermodynamics of solutions (ii) Concept of phase transition (iii) Phase separation (iv) Kinetics of the transition	2.1-3.6 (T1)	Students will be able to understand the concept of phase transition
Theory of Polymers (8-16)	(i) Elasticity of Polymers (ii) Random Walk model of polymers (iii) Rubber elasticity	3.1-3.5 (T1) 5.1-5.5 (T2)	Students will develop an understanding of the polymers and their elastic response
Theory of Colloids (17-23)	(i) What are Colloids? (ii) Flow of colloids: Stokes' law (iii) Brownian motion & Einstein equation (iv) Forces between colloids	4.1-4.4 (T2)	Student will be able to know about this class of matter. They will learn about the forces and motions in colloids
Theory of Liquid crystals (24-30)	(i) Phases in liquid crystals (ii) The nematic to isotropic transition (iii) Topological defects in liquid crystals (iv) Polymer liquid crystals	5.1-5.3 (T1) 7.1-7.7 (T2)	Students will develop an understanding about the liquid crystals. During discussion student will be able to understand the defects in liquid crystals and their importance.
Theory of Gels (31-34)	(i) Introduction of Gels (ii) Classes of gels (iii) Theory of gelation	6.1-6.3 (T2)	Students will develop an understanding about the different class of gels
Brownian Motion and thermal fluctuations (35-40)	(i) Brownian motion of a free particle (ii) Brownian motion in a potential field (iii) Fluctuation-Dissipation theorem	7.1-7.5 (T1)	Students will develop an understanding about the diffusion of soft matter in solutions

5. Evaluation Criteria:

Evaluation Component	Duration	Weightage	Date and Time	Mode
Tutorial tests	25 min	15%	Announced 4 tests - best 3 will be considered for the final marks	CB
Presentation-1 & VIVA	20-25 min	20%	TBA	OB
Presentation-2 & VIVA	20-25 min	25%	TBA	OB
Comprehensive Examination	180 min	40%	08/12/26 (AN)	CB

6. Note:

1. Since the course is four-level, learning won't be confined to standard textbook-based instruction. The course involves evaluating research papers and presenting an understanding of the assigned papers.

2. The evaluation scheme may change with prior notice.
3. Each student is expected to attend all classes and participate in all evaluation components.
4. **Consultation hours:** To be announced during the lecture.
5. **Notices:** All notices concerning the course will be uploaded on the Nalanda-aws portal only. <https://nalanda-aws.bits-pilani.ac.in/>

7. Make-up:

1. Make-up will be granted for any missing evaluative components due to genuine reasons.
2. For a comprehensive examination, the request should be made to AUGSD in accordance with its guidelines.
3. The decision to grant make-up to the applicant will be decided by the course instructor.

Prof. Navin Singh
Instructor-in-Charge
PHY F416